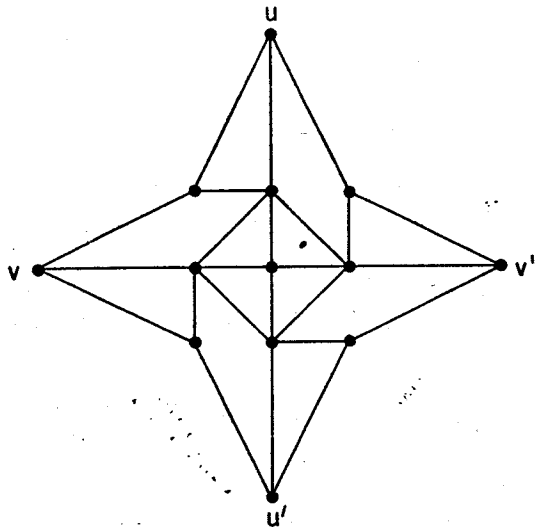


Fig. 2. Crossover H for Theorem 2.2.

(ii) For each edge $\{x, y\} \in A$, call its representation in the plane the $\{x, y\}$-*line*. To each such line which is "crossed" by other lines, add new points, one between each endpoint and the nearest crossing to it, and one between each pair of adjacent crossings.

(iii) Replace each crossing in the graph by a copy of graph H, identifying the outlets u and u' with the nearest new points on either side of the crossing on one of the lines involved, and identifying v and v' with the nearest new points on the other line.

(iv) For each $\{x, y\} \in A$, choose one endpoint as the *distinguished endpoint* and coalesce it with the nearest new point on the $\{x, y\}$-line. The edge between the other endpoint and *its* nearest new point on the $\{x, y\}$-line will be called the *operant edge* of the $\{x, y\}$-line.

This completes the construction of G'.

Suppose G' is 3-colorable and let $f: N' \to \{1, 2, 3\}$ be a valid 3-coloring. Then f restricted to $N \subseteq N'$ will be a valid 3-coloring of G. For suppose not. Then there would exist an $\{x, y\} \in A$ such that $f(x) = f(y)$. Consider the $\{x, y\}$-line in G' and assume without loss of generality that x is the distinguished endpoint for this line chosen in Step (iv). Then by (2.2A) all the new points on the $\{x, y\}$-line must have the same color as x. Thus both endpoints of the operant edge for that line have the same color, a contradiction.

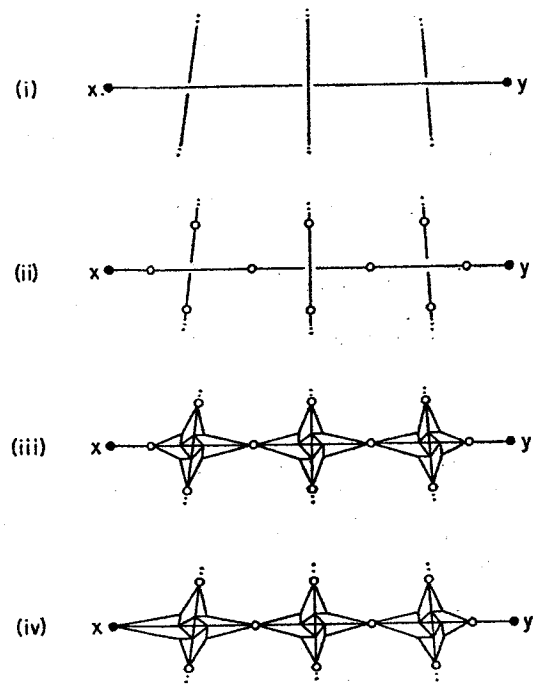


Fig. 3. Construction of G', as it affects the $\{x, y\}$-line.

Conversely, if $f: N \to \{1, 2, 3\}$ is a valid 3-coloring for G, it can be extended to 3-coloring for G' as follows: For each $\{x, y\} \in A$, color each new point on the $y\}$-line with color $f(x)$, where x is the distinguished endpoint of the line. This ures that all the operant edges of G' are validly colored. By (2.2B) this 3-coloring be extended to the interior nodes of the crossovers, thus yielding a valid 3-col- ng of G'.

Thus G' is 3-colorable if and only if G is, and the reduction is proved. □

orem 2.3. *Planar Graph 3-Colorability* α *Planar Graph 3-Colorability With Node ree At Most 4.*

f. The key to our construction will be the use of "node substitutes". Fig. 4(a) ws the 3-outlet node substitute H_3, with its first, second, and third outlet nodes elled. For $k \geqslant 4$, the k-outlet node substitute H_k is formed by adjoining to H_{k-1} opy of H_3 having its first outlet coinciding with outlet $k-1$ of H_{k-1}. The outlet